

SAE-Guided Trace Shortening for Reasoning Distillation

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Abstract

Longer teacher traces do not necessarily provide better supervision for smaller language models. We study **SAE-guided reasoning distillation**: identify length-associated features in a shared sparse autoencoder using within-problem contrasts between short and long answer-correct traces, steer teacher generation, and distill the resulting outputs. In an exploratory GSM8K study with a Qwen2.5-7B-Instruct teacher and Qwen2.5-1.5B-Instruct student, selected short-feature enhancement reduces teacher output length by 13.27%. Students trained on these outputs generate approximately 11% fewer tokens than students trained on unmodified outputs. Accuracy differences remain statistically inconclusive under equal-example and approximately equal-target-token budgets. The results support feature-guided length control and its downstream transfer, without establishing improved student accuracy.

1 Introduction

Chain-of-thought distillation uses teacher reasoning as supervision for smaller language models [Wei et al., 2022, Hsieh et al., 2023]. Yet longer traces need not be more useful: additional explanations may increase supervision volume without improving student learning [Li et al., 2025a]. The challenge is to produce concise teaching traces while retaining the information students need.

Existing approaches modify either completed traces or the generation process. LiteCoT rewrites teacher solutions through difficulty-aware prompting [Wu et al., 2025], while Compress-Distill illustrates that compression can reduce costs without improving accuracy [Griot et al., 2026]. ASC instead steers teacher activations using verbose-concise pairs [Azizi et al., 2026], and steering vector distillation shows that induced teacher behaviors can transfer through output-based training [Blank et al., 2026]. These precedents motivate a specific question: *can features associated with naturally short solutions guide teachers toward concise, useful student supervision?*

We investigate this question through **SAE-guided reasoning distillation**. We fit a shared sparse autoencoder (SAE) to pooled teacher activations and compare feature activity between short and long answer-correct solutions to the same problems [Cunningham et al., 2023, Gao et al., 2024]. This comparison holds teacher identity, problem content, and final-answer correctness fixed. Selected decoder directions then guide generation, and complete, answer-verified outputs supervise student fine-tuning. Our evaluated intervention enhances short-associated directions; these associations do not identify specific reasoning operations or certify redundant content.

On GSM8K [Cobbe et al., 2021], we compare SAE-guided supervision with unmodified generation, matched random interventions, and naturally short traces under equal-example and approximately equal-target-token budgets. The study connects within-problem feature discovery, generation-time intervention, and controlled student evaluation, separating verbosity control from supervision quality.

The selected intervention shortens teacher outputs and transfers shorter-output behavior to students, while its accuracy effects remain inconclusive. This makes two further distinctions central to the study: whether SAE control offers value beyond ordinary prompting or dense steering,

and whether its length association reflects reasoning content or answer-format cues. We therefore specify additional baseline comparisons and a falsification-oriented SAE analysis, clearly separating their proposed designs from completed results. The expanded evaluation links observable feature associations, measured changes under intervention, and the usefulness of the resulting supervision.

2 Related Work

Reasoning supervision and trace selection. Chain-of-thought reasoning exposes intermediate steps that can serve as supervision for smaller models [Wei et al., 2022, Hsieh et al., 2023]. However, stronger teachers and longer traces do not consistently improve small-student performance; mixing reasoning of different complexity can be more effective [Li et al., 2025a]. This motivates comparisons among random-correct, shortest-correct, and mixed-length supervision from a common candidate pool. Our within-problem design fixes teacher identity, problem content, and final-answer correctness, while separately accounting for supervision volume. It investigates the usefulness of generated sequences through SFT, without teacher-logit matching.

Compression before distillation. LiteCoT uses difficulty-aware prompting to rewrite teacher solutions into concise traces [Wu et al., 2025]. Compress-Distill studies post-hoc compression at multiple student scales and reports lower training and inference costs alongside an accuracy trade-off [Griot et al., 2026]. These methods establish textual compression as a direct comparator for constructing student supervision. Our framework modifies activations while the teacher generates a solution, but this implementation difference alone does not establish better teaching data. Comparisons must account for reference-trace generation, rewriting calls, usable-answer coverage, and student training budgets. An ordinary shortening prompt is also distinct from a full difficulty-aware rewriting pipeline and should be evaluated under its own name.

Activation-based reasoning control. ASC learns a steering vector from verbose–concise pairs to shorten chain-of-thought generation [Azizi et al., 2026]. Its published formulation uses a length-normalized contrastive energy objective and a KL trust-region constraint, making it distinct from a simple difference-of-means baseline. Other work extracts SAE features to steer reasoning and also studies an SAE-free alternative [Li et al., 2025b], or identifies strategy-associated features through keyword screening followed by intervention-based ranking [Fang et al., 2026]. Our focus is how naturally occurring length contrasts inform the generation of student training data. Dense directions, concise prompting, and norm-matched random feature sets test whether an SAE dictionary provides incremental value for this objective. Because our selected features have high activation near answer markers, an answer-format direction is an additional mechanistic control specific to the present evidence.

Sparse features, interpretation, and transfer. SAEs provide sparse decompositions of model activations, with TopK architectures controlling the number of active features [Cunningham et al., 2023, Gao et al., 2024]. Such decompositions do not guarantee that individual features correspond to reasoning operations. Ma et al. [2026] demonstrate that contrastively identified candidates can respond to lexical injections and non-reasoning counterexamples, motivating tests of semantic invariance and alternative explanations. We therefore separate observed length associations, intervention-induced changes to model states, changes to generated content, and effects on student learning. Steering vector distillation already demonstrates behavioral transfer through outputs from steered teachers [Blank et al., 2026]; the teacher-intervention-to-student-training pipeline itself is

not our novelty claim. Our proposed baseline and SAE analyses instead test what the selected directions control and whether that control provides useful supervision beyond simpler ways of shortening responses.

3 Experiments

3.1 Setup and Natural-Length Comparison

We use Qwen2.5-7B-Instruct as the teacher and Qwen2.5-1.5B-Instruct as the student, with 881 fixed GSM8K training problems [Cobbe et al., 2021]. Students undergo completion-only LoRA SFT over three seeds and are evaluated by final-answer accuracy and generated tokens on 1,269 fixed test problems, using greedy decoding with a 512-token cap. The primary layer-17, TopK-64 SAE identifies eight short-associated and eight long-associated features. Appendix A provides sampling, SAE, and training details.

In the initial equal-example comparison, natural-short supervision achieves 66.67% accuracy versus 64.20% for natural-long supervision, below the independently evaluated base student (67.45%). The 2.47-point difference is not significant after Holm correction ($p = 0.0852$). Unequal supervision-token volumes also prevent attributing this ordering to redundant content.

3.2 Teacher Intervention

Development screening selects enhancement of eight short-associated directions at relative perturbation magnitude $\rho = 0.30$, using random controls matched in feature count and perturbation norm. We fix the intervention before final student evaluation and generate four candidates per training problem under each condition. Table 1 includes all outputs, including incorrect and capped responses.

Generation condition	Correct (%)	Mean tokens	Cap hits
Unmodified	98.69	250.27	12
SAE short enhancement	98.89	217.07	1
Matched random	97.70	293.02	38

Table 1: Teacher generation on 881 problems, with 3,524 outputs per condition. Cap hits count responses reaching 512 tokens.

SAE enhancement shortens outputs by 13.27% relative to unmodified generation. Question-level bootstrap intervals give a mean length difference of -33.20 tokens (95% CI: $[-35.24, -31.22]$) and a correctness difference of $+0.20$ percentage points ($[-0.43, +0.79]$). These are post-hoc descriptive results; reused feature-analysis cohorts and the absence of a non-inferiority test preclude a claim of established accuracy preservation.

3.3 Downstream Distillation

For each generation condition, we deduplicate candidates and select each problem’s shortest answer-correct, non-capped solution. Intersecting support with the initial natural-short dataset leaves 878 shared problems. Natural-short supervision uses a different pipeline—quantile-band selection from 16 candidates rather than shortest-correct selection from four—so this comparison does not isolate steering. Four data conditions, two budgets, and three seeds yield 24 adapters. Token matching repeats complete solutions; it does not equalize optimizer steps or per-question weights.

Supervision	Equal examples	Equal target tokens
Unmodified	69.92	70.42
SAE short enhancement	70.84	70.34
Matched random	68.87	68.87
Natural short	66.93	66.96

Table 2: Mean student accuracy (%) over three SFT seeds. Target-token budgets are approximately matched. The base student scores 66.51% in this follow-up run.

Table 2 summarizes student accuracy. The SAE-minus-unmodified accuracy difference is +0.92 points under equal examples (95% CI: $[-1.34, +3.23]$; Holm-adjusted $p = 0.8432$) and -0.08 under equal tokens ($[-2.18, +1.97]$; $p = 0.9654$). Intervals use 10,000 crossed bootstrap replicates over seeds and paired evaluation problems; Holm correction covers six SAE-versus-control comparisons. Comparisons with unmodified and random generation remain inconclusive. SAE exceeds natural-short supervision under both budgets (adjusted $p = 0.0060$ and 0.0110), but the different sampling and selection pipelines confound attribution.

Student output length falls from 277.14 to 245.98 tokens under equal examples (11.24%) and from 276.55 to 245.31 under equal tokens (11.30%). The follow-up base averages 239.64 tokens, so shortening is relative to unmodified distillation, not the pretrained student. Its 66.51% accuracy also differs from the initial run’s 67.45%; the cause remains unresolved.

Scope. The results support transfer of shorter-output behavior, not verified accuracy gains or end-to-end compute savings. Evidence is limited to one teacher–student pair, GSM8K, one SAE seed, and reused analysis cohorts. Concise prompting, dense short–long steering, ASC, textual compression, and length-matched controls remain untested [Azizi et al., 2026, Wu et al., 2025]; the value of SAE decomposition beyond simpler alternatives remains open.

3.4 Baseline Comparison Design (Proposed)

The completed comparisons establish how the selected intervention behaves relative to its original controls. They leave open whether similar supervision can be obtained with simpler generation or selection methods. Table 3 specifies additional comparisons; no unreported performance values are assigned to them. The existing unmodified-generation student already uses shortest-correct selection from four candidates and must not be relabeled as random-correct. Likewise, the historical natural-short condition cannot substitute for a comparison using the same candidate pool and selection rule.

Selection, prompting, and internal controls. Random-correct and shortest-correct supervision would use the same unmodified pool, with random selection fixed before training. A concise prompt would request a complete solution with brief explanations while keeping the required final-answer format fixed. A dense baseline would average within-problem short–long activation differences with equal problem weighting at the same intervention layer. An answer-format baseline would instead use paired texts with the same mathematical steps and different answer headings, controlling aligned positions when estimating the direction. The latter is a targeted alternative explanation suggested by the feature-context evidence in Section 4.1, not an established mechanism. Additional random SAE sets would be drawn independently before evaluation, matching feature count and perturbation norm and reporting all sampled controls.

Comparison	Question addressed	Evidence status
Base student	Does the training pipeline improve on the initial student?	Reported
Unmodified + shortest-correct	Does steering add value with selection held fixed?	Reported
Matched random SAE	Is the chosen direction more useful than a matched perturbation?	One selected set reported
Natural short	How does the historical length-band pipeline compare?	Reported; selection differs
Random-correct, same pool	What does shortest-correct selection add to ordinary correct sampling?	Proposed
Concise prompt	Can explicit brevity instructions achieve comparable results?	Proposed
Dense short-long direction	Does an SAE decomposition add value over a direct activation contrast?	Proposed
Answer-format direction	Can answer formatting or an answer-transition correlate explain shortening?	Proposed
ASC	How does the method compare with learned activation compression?	Proposed
Text compression / DAP	How does internal control compare with rewriting teaching traces?	Proposed

Table 3: Baseline coverage and planned additions. Reported rows refer only to the completed exploratory study; proposed rows have no completed results in this manuscript. Additional independent random feature sets are also proposed.

Published compression methods. ASC would be implemented according to a specified published version, retaining its CES objective and KL constraint [Azizi et al., 2026]. A plain difference-of-means direction would remain a separate baseline. Text compression would rewrite the same teacher solutions before student SFT; a full difficulty-aware pipeline would be distinguished from a generic shortening prompt [Wu et al., 2025]. Any additional reference model, rewriting call, or data requirement would be disclosed and included in cost accounting. These methods would be evaluated with the same student rather than compared through their papers’ original benchmark scores.

Two complementary fairness questions. A fixed-candidate comparison would equalize the number of sampling opportunities, verification rules, and shortest-correct selection across generation methods. We would report coverage and failures before constructing a common-support student comparison, since intersections can hide problems on which a method fails. A separate fixed-generation-token comparison would assess usable-correct-trace yield under a common acquisition budget. These budgets answer different questions and would not be presented as simultaneously equal. Student experiments would retain separate equal-example and approximately equal-target-token analyses, documenting repeat counts and optimizer steps.

Length matching and calibration. Steering strengths and prompt variants would be calibrated on development problems to obtain comparable length ranges, with all tested settings disclosed. Accuracy-length curves would accompany comparisons at a chosen compression level; equal hidden-state norms alone do not imply equal behavioral effects. Within-problem length matching could provide a complementary conditional comparison, with its tolerance, residual imbalance, and retained support reported. Because length is itself affected by the method, post-hoc matching would not alone establish a causal content effect. Selection would use separate GSM8K training

problems, and final student evaluation would retain the fixed cohort. Repeated use of that cohort would remain explicit rather than being described as a newly untouched confirmation set.

Endpoints and interpretation. Primary downstream comparisons would report paired student accuracy with uncertainty, alongside student output length and actual training tokens. Teacher endpoints would include all-output correctness, cap hits, coverage, and usable correct traces per generated million tokens. End-to-end cost would include activation collection, SAE or direction fitting, candidate generation, verification, rewriting, and online intervention overhead. Both total cost and marginal cost with a reusable dictionary would be reported. An inconclusive accuracy difference would not establish non-inferiority, even if the method yields shorter outputs.

4 SAE Analysis

This section separates diagnostics already available from the completed study from proposed analyses of the intervention mechanism. The existing evidence establishes length associations and shorter generated outputs; it does not establish that the selected features encode redundant reasoning. The analyses below are intended to test that interpretation and to identify which aspects of the intervention merit downstream evaluation.

4.1 Existing Evidence: Answer-Marker Associations

The held-out token-context report for the primary SAE shows that the highest-activation examples of all eight selected short-associated features occur at **Answer** tokens. For example, feature F7339 has high-activation contexts such as a completed calculation followed by **Answer**:: analogous contexts occur in both short and long traces. These examples indicate an answer-format association, but do not measure how much of the feature’s total activation is explained by the marker. They motivate distinguishing a feature that tracks useful concise reasoning from one that tracks answer formatting or the transition into an answer segment.

A simple aggregation effect can also produce a short–long association. If a marker appears once per trace and contributes approximately constant activation A_j , a feature’s whole-trace mean can behave as

$$\bar{z}_j(\tau) = \frac{1}{L_\tau} \sum_{t=1}^{L_\tau} z_{t,j} \approx \frac{A_j}{L_\tau}. \quad (1)$$

Thus, token-normalized activation can correlate with shortness even when activation at the marker is unchanged. This is a possible explanation, not a fitted result. The existing first-64-token direction check does not by itself resolve it, because some traces may already contain an answer marker within that window.

4.2 Separating Length, Position, and Lexical Cues

We propose decomposing feature activation into reasoning-body, answer-marker, and answer-suffix regions, reporting both region-specific means and contributions to total activation. The analysis would compare short and long traces within the same problem and, where supported, at matched token identities and absolute positions. A before/after comparison excluding answer regions would test whether the observed association survives removal of its most evident correlate. Regressing whole-trace means against inverse length provides a complementary diagnostic of Equation 1; it does not replace the within-problem analysis.

A controlled text battery would preserve mathematical steps while changing wording or answer headings, preserve formatting while changing calculation correctness, and inject associated tokens into non-reasoning text [Ma et al., 2026]. Step and answer checks are needed to verify the intended manipulations. Because deletion and rewriting shift token positions and subsequent states, analysis would align corresponding spans and include position-matched controls rather than compare only whole-trace averages. Passing these tests would strengthen a semantic interpretation; failing them could still leave a useful formatting-control direction.

Early-prefix prediction would use only information available after 32 or 64 generated tokens to predict remaining length or subsequent entry into the answer segment. Evaluation would group by problem and report the fraction of traces ending before each landmark. Relative positions computed using eventual trace length and retrospective end-of-trace masks would be restricted to offline diagnostics, avoiding future information in an online predictor.

4.3 Target Engagement and Direction Specificity

The selected intervention adds a constant composite decoder direction scaled to the current hidden-state norm. To determine what it actually changes, we propose encoding the same states before and after the implemented perturbation:

$$z_t = \text{Enc}(h_t), \quad z'_t = \text{Enc}(h_t + \delta_t). \quad (2)$$

Readback must reproduce the deployed layer, normalization, feature aggregation, and numerical precision. Comparing independently generated texts would confound the perturbation with changed token histories.

Measurements would include the signed change in selected features, the magnitude of changes in unselected features, turnover of the TopK active set, and actual perturbation-to-state norm. The true next-token distribution after the remaining model layers would provide KL divergence and probabilities of answer markers and end-of-sequence tokens. These quantities are especially relevant at $\rho = 0.30$: matching hidden-state perturbation norms does not ensure comparable distribution shifts or feature specificity. A fixed decoder-direction injection and activation-gated feature suppression would remain separately named interventions.

4.4 Feature Redundancy, Timing, and Stability

Composition of the eight-feature direction. We propose inspecting pairwise decoder cosine similarity, activation correlations, and the singular-value spectrum of the selected directions. Shared answer-marker contexts make it important to determine whether eight features provide several independent effects or largely overlapping directions. Single-feature, top-1/2/4/8, and leave-one-out interventions would hold the total perturbation norm fixed. Directions would be ranked on development data before downstream evaluation. Larger sets would require additional features passing the same discovery rule, rather than relaxing selection thresholds to fill a requested set size.

When shortening occurs. Matched continuations from a shared natural prefix would compare enhancement, zero intervention, sign reversal, and multiple random controls with independent caches. Early-only, delayed, and answer-marker-triggered windows would localize the effect using only information available during generation. Outputs would be decomposed into answer-preceding steps, repeated explanations, and answer-following text. A reduction concentrated after the answer would support a different interpretation from preserved calculations with fewer repeated derivations. Response correctness and usable-answer coverage would accompany every length comparison.

Dictionary quality and reproducibility. The six existing SAEs support a first comparison across layers and sparsities without training a new grid. Their reconstruction and activation-coverage metrics are reported in Appendix B.1. Additional seeds for the primary setting would test whether the feature structure and intervention effects recur under dictionary retraining. Matching would use decoder geometry and activation profiles on independent questions, allowing feature splitting; feature IDs and the number of significant candidates are not measures of semantic stability.

4.5 Connecting Feature Changes to Student Learning

The decisive endpoint remains student performance after training on the generated data. We propose comparing SAE, dense or ASC, and prompt-based traces at similar length distributions under the training controls in Section 3.4. Method-blinded annotation would distinguish calculations, explanations, verification, repeated content, and answer formatting on paired problems. Student negative log-likelihood over these segments could provide a low-cost diagnostic of learnability, but lower likelihood loss would not establish a subsequent generalization gain. The multi-answer extension in Appendix A.4 would additionally separate multiple distinct texts from repeated exposure to the same text, subject to its unequal-token limitation. These analyses would connect a measured change in teacher behavior to the content and learning effects of the resulting supervision, without presupposing an accuracy benefit.

References

- Seyedarmin Azizi, Erfan Baghaei Potraghloo, Souvik Kundu, and Massoud Pedram. Activation steering for chain-of-thought compression. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 36676–36687. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.findings-acl.1828. URL <https://aclanthology.org/2026.findings-acl.1828/>.
- Camila Blank, Agam Bhatia, Senthorean Rajamanoharan, Arthur Conmy, and Neel Nanda. Subliminal learning is steering vector distillation, 2026. URL <https://arxiv.org/abs/2606.00995>.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems, 2021. URL <https://arxiv.org/abs/2110.14168>.
- Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models, 2023. URL <https://arxiv.org/abs/2309.08600>.
- Yi Fang, Wenjie Wang, Mingfeng Xue, Boyi Deng, Fengli Xu, Dayiheng Liu, and Fuli Feng. Controllable LLM reasoning via sparse autoencoder-based steering. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 21290–21305. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.acl-long.974. URL <https://aclanthology.org/2026.acl-long.974/>.
- Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders, 2024. URL <https://arxiv.org/abs/2406.04093>.
- Maxime Griot, Paul Steven Scotti, and Tanishq Mathew Abraham. Compress-distill: Reasoning trace compression for efficient knowledge distillation, 2026. URL <https://arxiv.org/abs/2606.05988>.
- Cheng-Yu Hsieh, Chun-Liang Li, Chih-Kuan Yeh, Hootan Nakhost, Yasuhisa Fujii, Alexander Ratner, Ranjay Krishna, Chen-Yu Lee, and Tomas Pfister. Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes, 2023. URL <https://arxiv.org/abs/2305.02301>.
- Yuetai Li, Xiang Yue, Zhangchen Xu, Fengqing Jiang, Luyao Niu, Bill Yuchen Lin, Bhaskar Ramasubramanian, and Radha Poovendran. Small models struggle to learn from strong reasoners, 2025a. URL <https://arxiv.org/abs/2502.12143>.
- Zihao Li, Xu Wang, Yuzhe Yang, Ziyu Yao, Haoyi Xiong, and Mengnan Du. Feature extraction and steering for enhanced chain-of-thought reasoning in language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 10893–10913. Association for Computational Linguistics, 2025b. doi: 10.18653/v1/2025.emnlp-main.552. URL <https://aclanthology.org/2025.emnlp-main.552/>.
- George Ma, Zhongyuan Liang, Irene Y. Chen, and Somayeh Sojoudi. Do sparse autoencoders identify reasoning features in language models?, 2026. URL <https://arxiv.org/abs/2601.05679>.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models, 2022. URL <https://arxiv.org/abs/2201.11903>.
- Yifan Wu, Jingze Shi, Bingheng Wu, Jiayi Zhang, Xiaotian Lin, Nan Tang, and Yuyu Luo. Concise reasoning, big gains: Pruning long reasoning trace with difficulty-aware prompting, 2025. URL <https://arxiv.org/abs/2505.19716>.

A Experimental Details and Scope

A.1 Sampling and Feature Discovery

Teacher sampling uses temperature 0.7, top- p 0.95, and a 512-token generation cap. The initial pool contains 16 candidates for each of 881 problems (14,096 traces). Shared TopK SAEs are fit on pooled residual activations from raw candidates regardless of correctness; only answer-correct traces enter short-long feature comparisons. Question-level splits contain 617 training, 132 development, and 132 test problems. The SAE grid covers layers 10, 17, and 23; TopK sparsities 32 and 64; expansion factor 8; and 1,500 training steps, using seed 17. Registered screening of the primary layer-17, TopK-64 dictionary yields eight features per length group.

Natural-length comparisons require at least four distinct, answer-correct, non-capped candidates per problem. Short and long representatives come from the bottom and top 20% length bands; medium representatives are near the median. All conditions retain the same 881 problems. Medium supervision achieves 66.33% seed-mean accuracy, between short (66.67%) and long (64.20%). Holm correction covers three registered natural-length comparisons. Final-answer correctness does not certify intermediate steps.

A.2 Intervention Screening

We screen short-feature enhancement and joint short-enhancement/long-direction subtraction at $\rho \in \{0.05, 0.15, 0.30\}$. Each has a control matched in decoder-column count and actual perturbation norm, excluding target features from random sets. Including unmodified generation gives 13 conditions. The joint variant uses a fixed composite direction, not activation-gated suppression. Interventions act at the layer-17 post-block residual position predicting each next token, beginning with the first generated token. Conditions share prompts and per-question random uniforms but maintain independent attention caches.

Selection uses 64 development problems. The shortest eligible condition must shorten mean length by at least 5%, lose no more than five accuracy points, have a paired 95% bootstrap length-difference interval below zero, and be shorter on average than its matched random control. A subsequent 64-problem check gives 246.44 versus 209.23 mean tokens, with unmodified and selected generation both correct on 64/64 problems. Both cohorts were previously used in SAE analysis; the accuracy threshold is not a non-inferiority test. The full teacher pool contains 10,572 outputs; its bootstrap intervals resample problems while retaining all four candidates together.

A.3 Student Training and Budget Accounting

All conditions start from the same pretrained student. LoRA uses rank 4, scaling factor 16, dropout 0.05, and all linear modules. Completion-only SFT runs for one epoch over each constructed dataset, with batch size 4, gradient accumulation 1, learning rate 2×10^{-5} , warmup ratio 0.03, BF16, maximum sequence length of 2,048 tokens without packing. Student seeds are 17, 42, and 73; these vary SFT, not SAE discovery. Evaluation uses `test` [50:1319]; the first 50 test problems are reserved for smoke checks. Final evaluation accuracy is not used to select interventions.

After deduplication and shortest-correct selection, unmodified, SAE, and random generation cover 881, 880, and 879 problems, respectively. Intersecting these with natural-short support leaves 878 problems. Equal-example datasets contain 878 records and require 220 optimizer steps. Repeating complete solutions brings target-token totals to 215,643–215,722 (Table 4). Random-control data, weights, and predictions are identical across budgets for each seed, not independent replications.

Supervision	Equal-example target tokens	Equal-token steps
Unmodified	188,540	252
SAE short enhancement	165,530	286
Matched random	215,643	220
Natural short	188,420	251

Table 4: Training-budget accounting. Every equal-example condition uses 220 steps; equal-token conditions have approximately matched token totals but different step counts.

A.4 Uncompleted Extension and Interpretation

A separately registered extension compares one, two, and four distinct correct responses per problem under SAE and unmodified generation on the same 878 problems. Repeating the shortest response two or four times controls record count, question exposure, and optimizer steps, but not target-token counts. Distinct text need not represent distinct mathematical solutions. No complete audited results are available, so no conclusions are drawn from partial runs.

Lexical and answer-format explanations remain possible: random directions control feature count and perturbation magnitude, not semantics or activation prevalence. End-to-end savings would also need to include SAE fitting, candidate generation, selection, and intervention overhead. The present evidence does not establish that long traces are generally redundant.

B SAE Diagnostics and Proposed Analysis Protocol

B.1 Existing Dictionary-Quality Measurements

Table 5 reproduces the completed SAE audit’s held-out reconstruction and activation-coverage measurements. The primary layer-17, TopK-64 SAE has explained variance 0.776 and approximately 0.18% of features inactive on the measured test support. The much higher inactive fractions at TopK 32 motivate inspecting training-set activity and learning curves before interpreting differences in the number of discovered features. Test inactivity is not equivalent to a feature never activating during training. These measurements neither establish semantic purity nor demonstrate that reconstruction quality predicts steering utility.

Layer	TopK	Test explained variance	Test-inactive features (%)
10	32	0.812	53.52
10	64	0.837	1.38
17	32	0.747	36.18
17	64	0.776	0.18
23	32	0.759	18.16
23	64	0.789	0.10

Table 5: Existing measurements from the six completed SAEs, all trained with seed 17. Inactivity is measured on the audit’s test support, not asserted over the full training distribution.

B.2 Region-Specific Activation and Counterexamples (Proposed)

The reported eight short-associated features are F7339, F10807, F17885, F27038, F22686, F11240, F24148, and F26001 in the primary dictionary. All have reported high-activation examples at **Answer**; their IDs are dictionary-specific. For each feature, a region-specific analysis would report activation mass in the answer marker and suffix, body-token means, marker occurrence counts, and activation per marker occurrence. Paired effect estimates before and after excluding these regions would be displayed with uncertainty and support counts. Global sums would be diagnostic only, because they also depend on trace length. Discovery-set nuisance adjustments would be frozen before application to separate questions.

Meaning-preserving rewrites would be checked for the same numerical dependencies and answer, while format-preserving incorrect variants would change a controlled calculation. Non-reasoning examples with injected marker tokens would test whether arithmetic content is necessary for activation. Results would include both successful and failed counterexamples rather than only the most interpretable feature examples. Deleting a word changes subsequent positions and states; aligned-span comparisons and matched control edits would therefore accompany lexical scrubbing.

For early-prefix prediction, models using lexical counts and absolute position would serve as controls for the SAE representation. Dense residual features and PCA features would be additional representation baselines under the same question-grouped splits and development tuning budget. Only traces still generating at a fixed landmark contribute to remaining-length prediction, with early-termination coverage reported separately. Prediction quality measures readable information; it does not establish that the predicted feature causes the outcome.

B.3 Readback and Geometric Controls (Proposed)

Let $\mathcal{F}$ denote the selected feature set and $\Delta z = z' - z$ the same-state readback difference. We would report signed changes in the selected features, $\|\Delta z_{\mathcal{F}}\|_2$, and $\|\Delta z_{\overline{\mathcal{F}}}\|_2$ separately. An off-target-to-target ratio would be accompanied by its component magnitudes and the fraction of states with a near-zero target change, avoiding interpretation driven by an unstable denominator. Because aggregate off-target norms depend on how many coordinates are included, per-coordinate distributions and matched-random controls would also be reported. The KL convention would be fixed as $D_{\text{KL}}(p_0 \| p_{\delta})$ using the actual next-token distributions after all remaining layers.

Decoder cosine similarities, activation correlations, and singular values would characterize redundancy within the selected set. A discovered answer-format subspace could further decompose the composite direction v into its projection and residual:

$$v_{\text{format}} = P_{\text{format}}v, \quad v_{\text{residual}} = (I - P_{\text{format}})v. \quad (3)$$

The subspace would be estimated on discovery data and tested on different questions. Interventions would disclose both component norms and any normalization multiplier; a negligible residual would not be amplified into a nominally equal-norm control. Persistence of an effect in the residual would test geometric separability from this estimated format subspace, not certify a pure reasoning mechanism.

B.4 Continuation, Robustness, and Student Checks (Proposed)

Continuation tests would clone caches from the same naturally sampled 32- or 64-token prefix and record prefixes that end too early or already contain an answer marker. Shared random uniforms reduce sampling variation but do not force identical paths after conditions diverge. Windowed or single-position interventions would evaluate when an effect arises; rules triggered by future trace length would not be used for online steering. Sign reversal need not produce a symmetric response, so dose curves, correctness, and coverage would be reported together.

Additional SAE seeds 42 and 73 at the primary setting would reuse a fixed corpus and activation-sampling budget. Feature matching would combine decoder geometry with activation profiles on separate questions and allow one-to-many correspondences. A question- or trace-balanced token-sampling sensitivity check would test whether longer traces influence dictionary fitting primarily through their greater token count. Reconstruction error would also be stratified by length, correctness, and position. None of these proposed retrainings is included in the current three-seed student result.

Method-blinded paired annotation would count preserved mathematical dependencies, repeated calculations, explanations, verification, and answer formatting. Student likelihood comparisons would separate body, calculation, and answer tokens and report segment lengths. The final test would remain SFT on matched-support datasets and paired evaluation, rather than treating likelihood or text similarity as teaching-utility labels. Suggested visual summaries are region-by-feature activation matrices, cleaned versus original paired effects, target/off-target dose curves, and student accuracy-length plots with uncertainty. These figures require the corresponding completed measurements; no synthetic results are supplied for them in this draft.